

Generation Interconnection System Impact Study Report

For Queue Project Q22-1985

Camden 138 kV
November 21, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1985. The Interconnection Customer proposes a 181.2 MW battery energy storage facility interconnecting at the Camden 138 kV substation in the Pennsylvania area. The request was received on January 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 714 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,887,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MURRAY / CAMDEN 138 kV	OAKLAND X / LA CROSSE 138 kV	149.2	208	100.0
GUILBEAU ELECTRIC / OAK HALL 138 kV	GUILBEAU ELECTRIC / MOUNT HOLLY 138 kV	145.9	333	5.5
ASH ST STA 223 / TOMBSTONE 345 kV	MARCUS HOOK ENERGY LP / TOMBSTONE 500 kV Line	133.0	1,413	7.6
CROMBY SUBSTATION / ASH STREET STATION 223 345 kV line	MARCUS HOOK ENERGY LP / TOMBSTONE 500 kV line	128.2	1,212	6.6
CROMBY / OWATONNA 345 kV	CURECANTI / MOSER GENERATING STA SUBSTATION 345 kV	121.8	1,147	5.7
GREENBACK-FAIROAKS / LA PARGUERA 6603 138 kV	WOODLYNNE / RHOME 138 kV line	117.7	240	9.2
ELK RIV / FLANNIGAN SUBSTATION 345 kV	TOMBSTONE / BIXBY HILL SUBSTATION 500 kV	116.4	1,065	5.5
TOMBSTONE 500 kV/345 kV Transformer Bank	MARCUS HOOK ENERGY LP / TOMBSTONE 500 kV	112.9	1,608	9.0
RHOME / WOODLYNE 138 kV	GREENBACK-FAIROAKS / SOUTH BEACH 138 kV	112.8	298	7.6
BURHOLME / GORDON EVANS NERGY CENTER 138 kV line	OAKLAND X / LA CROSSE 138 kV	112.6	205	49.3
HATHAWAY ST / DORCHESTER 138 kV L	TOMBSTONE / MARCUS HOOK ENERGY LP 500 kV Line	111.9	349	5.0
GREENBACK-FAIROAKS / SOUTH BEACH 138 kV	WOODLYNNE / RHOME 138 kV line	111.7	298	9.2
CORNELIUS DELIVERY NO 1 / BERREND C 345 kV	MOSER GENERATING STA / OWATONNA 345 kV L	111.6	1,223	5.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CLEAR SPRINGS 500/345 kV Transformer Bank	HILLS CHAPEL / BURNS 345 kV Line	111.4	1,702	8.4
HURRICANE CREEK / WOODLYNNE 138 kV	GREENBACK-FAIROAKS / S BEACH 138 kV	111.3	188	6.0
CECIL FIELD / DORCHESTER 138 kV	TOMBSTONE / MARCUS HOOK ENERGY LP 500 kV L	110.8	370	5.0
FRUITLAND / E BENCH 138 kV Line	MARCUS HOOK ENERGY LP / TOMBSTONE 500 kV L	110.5	229	5.0
CHILLIWICK / SEA ISLAND 138 kV L	SUPER MAX SUBSTATION / GREEN COUNTRY 138 kV line	110.3	558	7.9
PLYMOUTH MEETING / CECIL FIELD 138 kV	MARCUS HOOK ENERGY LP / TOMBSTONE SUBSTATION 500 kV	110.1	384	5.0
BARBADOES / PLYMOUTH MEETING 138 kV	TOMBSTONE / MARCUS HOOK ENERGY LP 500 kV	109.3	402	5.0
26TH AND F / CHILLIWICK 138 kV	GREEN COUNTRY / SUPER MAX 138 kV	109.1	596	7.9
WALNUT / CLEAR SPRINGS 345 kV	WALLENPAUPACK 500/345 kV Transformer	108.7	1,676	6.4
WAREHOUSE ELECTRIC / LIBERTY ELECTRIC POWER PLANT 138 kV	SEA ISLAND / 6TH ST 138 kV	108.4	427	7.5
SEA ISLAND / 6TH STREET 138 kV L	GUILBEAU ELECTRIC 345/138kV Transformer Bank	108.3	353	6.6
BRIDEPORT SUBSTATION / MARCUS HOOK ENERGY LP 138 kV	CASTLETON / PLAINFIELD 138 kV	108.1	479	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MURRAY / CAMDEN 138 kV	Rebuild line with higher-capacity conductor	\$413,000	100% (\$413,000)
OAK HALL / GUILBEAU ELECTRIC 138 kV L	Rebuild line with higher-capacity conductor	\$5,112,000	50% (\$2,557,000)
ASH ST STA 223 / TOMBSTONE 345 kV	Rebuild line with higher-capacity conductor	\$28,634,000	2% (\$512,000)
CROMBY / ASH ST TA 223 345 kV	Rebuild line with higher-capacity conductor	\$20,692,000	2% (\$360,000)
CROMBY / OWATONNA 345 kV	Rebuild line with higher-capacity conductor	\$20,959,000	2% (\$484,000)
GREENBACK-FAIROAKS / LA PARGUERA 6603 138 kV	Rebuild line with higher-capacity conductor	\$345,000	29% (\$100,000)
ELK RIV / FLANNIGAN SUBSTATION 345 kV	Rebuild line with higher-capacity conductor	\$41,534,000	2% (\$648,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TOMBSTONE 500/345kV Transformer	Replace transformer with higher-rated unit	\$33,403,000	2% (\$768,000)
WOODLYNE / RHOME 138 kV	Rebuild line with higher-capacity conductor	\$846,000	25% (\$210,000)
BURHOLME / GORDON EVANS NERGY CENTER 138 kV line	Rebuild line with higher-capacity conductor	\$1,444,000	94% (\$1,357,000)
DORCHESTER / HATHAWAY ST 138 kV	Rebuild line with higher-capacity conductor	\$1,617,000	30% (\$486,000)
GREENBACK-FAIROAKS / SOUTH BEACH 138 kV	Rebuild line with higher-capacity conductor	\$6,841,000	29% (\$1,989,000)
CORNELIUS DELIVERY NO 1 / BERREND A C 345 kV	Rebuild line with higher-capacity conductor	\$26,576,000	2% (\$458,000)
CLEAR SPRINGS 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$31,994,000	2% (\$552,000)
HURRICANE CREEK / WOODLYNNE 138 kV	Rebuild line with higher-capacity conductor	\$2,908,000	31% (\$893,000)
CECIL FIELD / DORCHESTER 138 kV	Rebuild line with higher-capacity conductor	\$1,713,000	30% (\$515,000)
FRUITLAND / E BENCH 138 kV	Rebuild line with higher-capacity conductor	\$1,941,000	30% (\$584,000)
CHILLIWICK / SEA ISLAND 138 kV L	Rebuild line with higher-capacity conductor	\$8,361,000	25% (\$2,094,000)
PLYMOUTH MEETING / CECIL FIELD 138 kV	Rebuild line with higher-capacity conductor	\$1,906,000	30% (\$573,000)
BARBADOES / PLYMOUTH MEETING 138 kV	Reconductor line	\$1,975,000	30% (\$594,000)
26TH AND F / CHILLIWICK 138 kV	Reconductor line	\$640,000	25% (\$160,000)
WALNUT / CLEAR SPRINGS 345 kV	Reconductor line	\$10,192,000	2% (\$174,000)
WAREHOUSE ELECTRIC / LIBERTY ELECTRIC POWER PLANT 138 kV	Reconductor line	\$459,000	15% (\$69,000)
SEA ISLAND SUBSTATION / 6TH ST SUBSTATION 138 kV	Reconductor line	\$1,059,000	28% (\$298,000)
BRIDEPORT SUBSTATION / MARCUS HOOK ENERGY LP 138 kV	Reconductor line	\$319,000	12% (\$39,000)
	Total allocated to project		\$16,887,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.